

Final Exam Instructions

Be verbose. Explain clearly your reasoning, methods, and results in your written work.

No code is necessary, but including it in your answer could result in partial credit.

Written answers are worth the amounts stated.

Total points available: 120. The exam is graded out of 100 -- There are 20 extra credit points available.

General Guidelines

1. **Format:** Answers should be formatted as a PDF. You may convert your Python notebook, if you use one, directly to PDF. Alternatively, use an editor like Word and print to PDF.
2. **Restate the question:** Include the question number and restate each question before your answer.
3. **Submission:** When finished, upload your PDF to Canvas along with your code (optional).
4. **Repository:** Do not check code or answers into your repository until after the exam is completed by all.
5. **Data:** Data for problems are available in the repository named by the question number.
6. **Permitted resources:**
 - You may use your notes and the internet **for coding syntax help only.**
 - **You may not use an LLM** – turn off any coding helper you may have installed. Using an LLM for guidance even if not answering the question directly is not allowed.
 - **You may not work with other students** – all work must be your own.
7. **Honor Code:** All students will be held to the **Duke Community Standard.**

Duke's Community Standard:

Duke University is a community dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, respect, and accountability. Citizens of this community commit to reflect upon and uphold these principles in all academic and non-academic endeavors, and to protect and promote a culture of integrity.

To uphold the Duke Community Standard:

1. I will not lie, cheat, or steal in my academic endeavors,
2. I will conduct myself honorably in all my endeavors; and
3. I will act if the Standard is compromised.

Final Exam Questions

1. (25 pts)

Using **problem1.csv**:

The data set contains simulated 1 day returns based on a model for returns. You want to price options based on this assumption. The series has been adjusted so that it maintains the risk neutral assumption on price growth.

- Current Stock Price - \$100
- 1 Day to Maturity
- Annual Risk Free Rate - 4%
- 255 trading days per year
- The Stock pays no dividend

- a. What is the price of a put and a call when the Strike $\in [99,100,101]$ (5 pts)
 - b. Graph the implied volatility of call option prices using BSM along the range of strikes between [98,102] (5 pts)
 - c. Discuss: Theory would say this should be a flat line. Why is it not? (15 pts)
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2. (75 pts)

Using **problem2.csv**:

You have 3 assets to invest in. 6 month simulated returns for each asset are in **problem2.csv. You plan to hold the assets for 6 months. The annual risk free rate is 4% (arithmetic annual compounding).**

Assume a budget constraint where the sum of weights = 1 for all optimizations

- a. What is the holding period Sharpe Ratio of each asset? (5 pts)
- b. Find the maximum Sharpe Ratio portfolio. Weights can be in $[-1,1]$ (5 pts)
- c. Explain the weights. Consider the correlations and Sharpe Ratios of the assets. (15 pts)
- d. Now use the simulated values, and Expected Shortfall as the risk measure to find the optimal portfolio for $(E(r) - r_f) / ES$. (10 pts)
- e. Discuss the weights. Why do they differ from the max Sharpe Ratio weights? (15 pts)

- f. Calculate the risk parity portfolio based on the simulation and ES. Report the weights and the Contribution to ES for each asset. (10 pts)
- g. Calculate the return and risk ex-ante attribution of the portfolio from part d. Compare to the return and risk ex-ante attribution of the risk parity portfolio. Discuss. (15 pts)
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3. (20 pts)

What conditions are necessary for a risk parity portfolio and a maximum Sharpe Ratio portfolio to be the same? Show it using an example with 3 assets. (20 pts)